VOUCH

Vouch for every agent decision.

9 agents · 4 frameworks · 2 LLM providers · 1 chat room · 1 signed receipt

Band of Agents Hackathon · Track 3 — Regulated & High-Stakes Workflows

`Live demo · vouch.apohara.dev`
`Source · github.com/SuarezPM/apohara-themis`

The problem

Regulated procurement needs receipts, not summaries

Buyer-side procurement — vendor onboarding, financial risk scoring, statutory compliance, audit sign-off — is a multi-agent workflow by construction. Today the agents are human. Tomorrow they will be AI. The gap is the receipt: an auditor must be able to verify, offline, that a decision was made, by whom, on what data, and why.

DORA Art. 17	72-hour incident reporting clock
EU AI Act Art. 12	Record-keeping for high-risk AI systems
NIST AI RMF	Govern / Map / Measure / Manage trace
OWASP Agentic 2026	ASI01–ASI10 threat model

Multi-framework compliance is the moat — one Evidence Packet, four regulators satisfied by construction.

Sponsor stack

Band · AI/ML API · Featherless — three load-bearing integrations

Band	Multi-agent coordination substrate; chat-room IS the audit trail (9 agents, 2 accounts)
AI/ML API	Multimodal reasoning for 5 of 9 agents × 4 models (GPT-5.4, Haiku 4.5, Sonnet 4.6, Opus 4.7)
Featherless	Open-weight specialist reasoning for 3 of 9 agents × 3 models (Llama-3.3-70B, Qwen3-Coder-30B, DeepSeek-V3)

Per-agent cost is visible in the live demo UI's cost panel. Graceful degradation between providers; no single-vendor lock-in; no consensus trap (3 distinct Featherless lineages).

9-agent court architecture

Single Band chat room (vouch-procurement-court) · 4 frameworks · 2 LLM providers

#	Agent	Framework	Model	Sponsor
1	@Orchestrator	LangGraph	openai/gpt-5.4	AI/ML API
2	@IntakeAgent	CrewAI	claude-haiku-4-5	AI/ML API
3	@VendorResearcher	LangGraph	Llama-3.3-70B	Featherless
4	@FinanceRiskAnalyst	Pydantic AI	claude-sonnet-4-6	AI/ML API
5	@LegalPolicyChecker	CrewAI	Qwen3-Coder-30B	Featherless
6	@RedTeamAuditor	Anthropic SDK	claude-opus-4-7	AI/ML API
7	@ComplianceVeto	Pydantic AI	claude-haiku-4-5	AI/ML API (2nd account)
8	@EvidenceClerk	LangGraph	DeepSeek-V3-0324	Featherless
9	@ApprovalManager	CrewAI	claude-sonnet-4-6	AI/ML API

Orchestrator + state machine

9-state machine, LangGraph, every transition signed

IDLE	Await first procurement case
INTAKE	@IntakeAgent extracts ProcurementCase (Claude Haiku 4.5)
RESEARCH	@VendorResearcher resolves vendor profile (Llama-3.3-70B)
RISK	@FinanceRiskAnalyst scores RiskScore (Claude Sonnet 4.6)
POLICY	@LegalPolicyChecker cites statutes (Qwen3-Coder-30B)
AUDIT	@RedTeamAuditor adversarially audits (Claude Opus 4.7)
REDTEAM	@ComplianceVeto binding veto (Claude Haiku 4.5, 2nd account)
EVIDENCE	@EvidenceClerk seals EvidencePacket (DeepSeek-V3)
DECISION / DONE	@ApprovalManager renders DecisionMemo + C2PA PDF + human sign-off

Every state transition emits thenvoi_send_event with message_type='thought'. The transcript IS the audit trail — signed and embedded in the Evidence Packet.

Court flow

Intake → Research → Risk → Policy → Audit → Veto → Evidence → Approval

```
case-WAYNE-2026-0173
@Orchestrator → @IntakeAgent → @VendorResearcher
→ @FinanceRiskAnalyst → @LegalPolicyChecker
→ @RedTeamAuditor → @ComplianceVeto
→ @EvidenceClerk → @ApprovalManager

@ApprovalManager → human sign-off → vouched: true
```

Every arrow is a real @mention in a live Band chat room. Every node is a real signed event in the BLAKE3 chain.

Evidence Layer (the VOUCH moat)

Rust workspace: vouch-chain · vouch-evidence · vouch-gate · vouch-receipt · vouch-aibom · vouch-compliance

Layer	Primitive	What it does
`vouch-chain`	BLAKE3	Hash chain over every agent decision, sequence-monotonic
`vouch-evidence`	Ed25519	Per-tenant signing; per-entry signature
`vouch-evidence`	RFC 3161	Trusted timestamp (FreeTSA), cert chain validated
`vouch-evidence`	Rekor v2	Transparency log inclusion proof
`vouch-receipt`	C2PA	Signed PDF manifest, vendor-neutral provenance
`vouch-aibom`	CycloneDX 1.6	AIBOM: every agent + every model lineage
`vouch-compliance`	DORA / EU AI Act / NIST AI RMF / OWASP Agentic	4 framework mappers
`vouch-gate`	BAAAR (5 conditions)	Deterministic post-LLM halt gate

vouch-verify CLI (offline)

Offline verification, no network, <30 s

```
$ vouch-verify fixtures/sample_packet.json

Reading SealedPacket: case-WAYNE-2026-0173
DSSE envelope: 9 signatures, 9/9 verified
BLAKE3 chain: 10 entries, chain root 7a3b4c2e...
Ed25519 sigs: 9/9 verified against per-agent pubkeys
RFC 3161: freetsa.org genTime=1718662400
Rekor v2: log_index=1287, included
EU AI Act: 8/8 Art. 12 fields populated
DORA: 3/3 Art. 9/10/17 fields populated
NIST AI RMF: 4/4 functions traced
OWASP Agentic: 10/10 threats assessed

✓ VERIFIED — packet is tamper-evident and timestamped
```

Live demo

vouch.apohara.dev — 3 panels, cold fetch <800 ms

Left panel	Live Band room transcript (SSE stream, auto-scroll, latest at bottom)
Top-right	Per-agent cost panel: AI/ML API $ + Featherless $ per call
Bottom-right	EU AI Act Art. 12 dashboard (8/8 ✓ on every approved packet)
Form	Submit procurement → agents deliberate → Evidence Packet → human sign-off → vouched=true

Demo arc (5 min): submit → 9-agent deliberation → Evidence Packet download → offline verify → human sign-off → vouched=true.

EU AI Act Art. 12 coverage

8/8 fields populated on every approved packet

EU AI Act Art. 12 field	Source	Status
start_time	RFC 3161 timestamp	✓
end_time	RFC 3161 timestamp	✓
reference_database	Band room id + chatroom_id	✓
input_data	ProcurementCase JSON envelope	✓
natural_person_id	Buyer + approver identifiers	✓
decision_id	case_id + sequence	✓
policy_version	vouch-compliance version	✓
hash_chain_prev	BLAKE3 prev_hash link	✓

The same Evidence Packet also populates DORA Art. 9/10/17 (3/3), NIST AI RMF (4/4 functions), OWASP Agentic 2026 (10/10), and the CycloneDX 1.6 AIBOM.

Cross-prize narrative

One submission, three prizes

Main Track 3	9-agent procurement court + cryptographic Evidence Layer + offline verifier (vouch-verify)
Best Use of AI/ML API	5/9 agents × 4 distinct models on the critical decision path
Best Use of Featherless	3/9 agents × 3 distinct open-weight models chosen by role

Two LLM providers, four frameworks, two Band accounts. No single-vendor lock-in. No consensus trap. The verb is *VOUCH*.

Main Track 3 — Regulated & High-Stakes

Prize: 1st / 2nd / 3rd

Artifact	9-agent procurement court on Band (LangGraph + CrewAI + Pydantic AI + Anthropic SDK)
Workflow	Regulated enterprise procurement: intake → research → risk → policy → audit → veto → evidence → approval
Compliance veto	Independent second Band account cannot be overridden by the Orchestrator
Evidence Layer	Ed25519 + BLAKE3 + RFC 3161 + Rekor v2 + C2PA — 9/9 signed per packet
Verifier	vouch-verify CLI · offline · <30s · no network · reproducible from the receipt
Coverage	EU AI Act Art. 12 8/8 · DORA Art. 9/10/17 3/3 · NIST AI RMF 4/4 · OWASP Agentic 10/10

One receipt per procurement case. Tamper-evident, timestamped, auditable without trusting VOUCH.

Best Use of AI/ML API

Prize: $1,000 cash + $1,000 credits

Coverage	5 of 9 agents on the critical decision path × 4 distinct models
@Orchestrator	openai/gpt-5.4 — 9-state machine routing, emits thenvoi_send_event on every transition
@IntakeAgent	claude-haiku-4-5 — structured extraction into typed ProcurementCase (9 fields)
@FinanceRiskAnalyst	claude-sonnet-4-6 — RiskScore with citation grounding; monotonic in amount_eur
@RedTeamAuditor	claude-opus-4-7 — adversarial audit, deterministic CRITICAL finding via Hypothesis
@ApprovalManager	claude-sonnet-4-6 — DecisionMemo + C2PA PDF + human sign-off gate
@ComplianceVeto (2nd Band account)	claude-haiku-4-5 — binding veto over Critical findings, escalation routing

Prompt-cache hit ≈95% across runs on Sonnet 4.6 — marginal cost stays low without sacrificing reasoning quality.

Best Use of Featherless AI

Prize: $500 cash + $300 + $100 credits

Coverage	3 of 9 agents × 3 distinct open-weight model lineages, each chosen by role
@VendorResearcher	meta-llama/Llama-3.3-70B-Instruct — sanctions / UBO / adverse-media resolution against fixture DB
@LegalPolicyChecker	Qwen/Qwen3-Coder-30B-A3B-Instruct — statutory citation grounding (EU Directive 2014/24/EU + GDPR + AMLD6 + DORA + SOX)
@EvidenceClerk	deepseek-ai/DeepSeek-V3-0324 — long-context evidence aggregation into typed EvidencePacket envelope
Wiring	Featherless OpenAI-compatible gateway (https://api.featherless.ai/v1) · graceful MockLlmProvider fallback in tests
Diversity	Three independent model lineages — no consensus trap, no two-agents-same-model deception

Open-weight specialists on every high-stakes research/policy/synthesis call — exactly the lane Featherless is built for.

Pricing tiers

Community · Pro · Enterprise

Community	Free · single tenant · 100 packets/mo · vouch-verify CLI
Pro	$499/mo · 5 tenants · unlimited packets · CycloneDX AIBOM
Enterprise	From $5k/mo · on-prem · KMS-issued keys · SSO + SAML · DPA

Roadmap

Phase A — D · what's next

Phase A	Production hardening: KMS-issued per-tenant keys (replace baked compile-time keys)
Phase B	Real OIDC identity for Sigstore Rekor publish (anchor in the public log)
Phase C	Self-hosted LLM endpoint (Qwen3-235B-A22B on AMD MI300X, $0/inference)
Phase D	IETF AAT format alignment (SHA-256 + ECDSA P-256, per the draft)

Ask

What Apohara VOUCH needs from the jury

1. **Main Track 3** — recognize the 9-agent procurement court + cryptographic Evidence Layer as the regulated-workflow artifact.
2. **Best Use of AI/ML API** — recognize 5/9 agents × 4 distinct models on the critical decision path.
3. **Best Use of Featherless** — recognize 3/9 agents × 3 distinct models as the open-weight specialist reasoning backbone.

The verb is VOUCH. Vouch for every agent decision.

Contact

Apohara VOUCH

Live demo	https://vouch.apohara.dev
Source	https://github.com/SuarezPM/apohara-themis
License	MIT
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Sponsor integrations	Band · AI/ML API · Featherless AI
Spec	docs/SPEC.md · docs/REFERENCES.md

Vouch for every agent decision.